Technical Manual: Drive Motor Roller Bearing Overhaul

Vibration Severity Diagnostics, BPFO Analysis & Replacement Protocol

ASSET ID: MOTOR-DRV-01 / BRG-301 **DOC REF:** MTM-BRG-301-REV2 **STANDARD:** ISO 10816-3 / DIN 616 **SECTOR:** Plant 1 Assembly

1. Vibration Spectral Envelope & Fault Signatures

Drive motor roller bearing **BRG-301** is monitored continuously by dual-axis piezoelectric accelerometers (horizontal and vertical channels):

- **Outer Race Defect Frequency (BPFO):** 108.4 Hz at 1480 rpm operational speed.
- **Normal Kurtosis:** 2.5 - 3.2 (Gaussian baseline vibration).
- **Alarm Kurtosis:** > 4.8 indicates subsurface spalling and raceway micro-pitting.
- **High-Frequency Demodulation:** Envelope peak > 16.0 dB confirms metallic impacts requiring bearing renewal.

2. Required Tools & Replacement Spares

Tool / Component	Specification / Model	Operational Purpose
Replacement Bearing	SKF 6208-2RS1/C3 Deep Groove	Primary shaft radial support
Bearing Puller	3-Arm Hydraulic Puller (50 kN)	Damage-free removal from rotor seat
Induction Heater	SKF TIH 030M Controlled Heater	Thermal expansion for shrink-fit mounting
Grease Compound	Polyurea Synthetic Grease (NLGI 2)	Low-friction elastohydrodynamic film

3. Step-by-Step Replacement Procedure

1. **Decouple Motor Shaft:** Disconnect elastomeric coupling to gearbox. Check shaft runout with dial test indicator (< 0.02 mm).
2. **Remove End-Bell:** Unbolt end-bell casing bolts and withdraw non-drive end housing.
3. **Extract Degraded Bearing:** Position puller arms behind inner ring shoulder only. Apply steady hydraulic pressure until extracted.
4. **Inspect Journal Seat:** Clean with mineral solvent and verify shaft diameter (40.002 - 40.011 mm).
5. **Mount New Bearing:** Heat new bearing to strictly 110°C on induction cone. Slide onto journal until seated against locating shoulder.

6. **Post-Installation Verification:** Rotate shaft manually to verify smooth rotation. Re-energize at 25% load; confirm vibration kurtosis returns below 3.0.